

Article

Auditable Alarm Diagnosis: Evidence-Role Separation and Risk-Bounded Selective Calibration for Industrial Fault Disposition

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Highlights

What are the main findings?

- A formal evidence contract makes current evidence, historical analogy, language explanation, and final disposition non-interchangeable and machine-auditable, binding every claim to a checkable gate.
- Audit-constrained selective calibration couples coverage maximization to one-sided bounds on dangerous and nuisance routing errors, attaining the highest admissible coverage among compared methods.

What are the implications of the main findings?

- Numerical disposition recovered many TEP faults missed by the language-model reader, and a single sealed confirmation reproduced both risk bounds, supporting a held-out effectiveness claim.
- Blinded two-expert validation confirmed the automated retrieval assay tracks human relevance (weighted kappa 0.684 expert–expert, 0.547 model–expert), grounding retrieval quality in human judgment.

Abstract

Industrial alarm diagnosis is difficult to audit when current measurements, historical analogies, generated explanations, and operational dispositions are treated as interchangeable. We introduce an evidence-role separation protocol and audit-constrained selective calibration (ACSC), which binds every released claim to a machine-checkable gate. ACSC maximizes automatic coverage among candidates whose one-sided Clopper–Pearson error bounds satisfy prespecified limits for fault-to-benign and normal-to-fault routing. On HAI 21.03, a label-free detector score preserves event ranking (AUROC 0.811, 49-event queue) and supports a calibrated abstaining disposition on an expanded 77-event frame (in-frame AUROC 0.902; specificity 22/32, both in-sample); structured dual-channel retrieval is not statistically distinguished from the full hybrid under a frozen assay, and blinded two-expert validation confirms the assay tracks human relevance (weighted kappa 0.684 expert–expert, 0.547 model–expert). On the Tennessee Eastman Process, a numerical router recognizes 330/400 fault runs (29/400 routed benign) on a locked stability check, recovering faults missed by the language-model reader; ACSC attains the highest admissible coverage among comparators while meeting both risk bounds, and a single sealed confirmation reproduces them (upper bounds 0.042 and 0.099) at 0.860 coverage. Prospective field validation and causal localization remain future work. Code and artifacts are prepared for release at https://github.com/lplsaiorg/AuditableAlarmDiagnosis_EvidenceRoleSeparation.

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1. Introduction

When an industrial alarm fires, operators must combine four kinds of information: current sensor measurements and detector scores, historical cases, language-model explanations, and the final operational disposition. In sensor-rich monitoring systems, alarm-management standards such as ANSI/ISA-18.2 and IEC 62682 require this decision process to be documented and auditable [1,2]. Each object nevertheless answers a different question: current measurements describe the event being assessed; a retrieved case is an analogy; generated text interprets provided material; and a disposition determines whether the event is treated as fault, benign, or unresolved. When one interface blends these roles, a plausible historical narrative can be mistaken for current evidence, or a fluent explanation can silently become the decision.

Several research traditions address parts of this problem. Time-series anomaly detectors provide scores or labels from current measurements. Case-based reasoning (CBR) retrieves prior situations [3], including for the Tennessee Eastman Process (TEP) [4]. Retrieval-augmented generation (RAG) gives language models external records at inference time [5]; recent work applies retrieval to time-series forecasting [6–8] and industrial monitoring or maintenance [9–11]. Selective classification and conformal risk control formalize abstention or risk-calibrated output under their respective assumptions [12,13]. Bayesian networks and graph neural networks instead use explicit or learned dependency structures for fault inference [14,15]. Table 1 summarizes the distinction relevant here.

Table 1. Related method families and the evidence-role boundary of this study. The comparison concerns task structure, not cross-study performance.

Method family	Typical evidence	Typical output	Boundary in this study
Bayesian or causal-graph diagnosis	Dependency or causal graph	Fault posterior or root cause	No causal or root-cause claim
Graph-neural industrial diagnosis	Learned equipment or signal graph	Class or anomaly score	Focus is evidence roles and audit lineage
Case-based reasoning	Historical cases	Nearest-case diagnosis	Historical outcome is restricted to analogy
Selective or conformal methods	Calibrated uncertainty	Prediction set or abstention	Deterministic selective gates only; no coverage guarantee
Industrial RAG	Documents or cases	Generated diagnosis	Reader output is immutable and cannot overwrite the numerical route

These methods do not by themselves define how evidence roles may interact in an auditable alarm workflow. Corrective retrieval, counterfactual reasoning, argument-based RAG, refusal, and stable RAG design address related reliability problems [16–20]. The remaining methodological gap is an executable non-interference protocol: each object needs a defined source, allowed operations, audit record, and claim gate. Provenance standards provide a basis for tracing derivation and revision [21], but provenance alone does not prevent a historical outcome from being asserted as a present fact.

We therefore ask: how can current detector evidence, historical analogies, language explanations, and selective dispositions be represented and evaluated as non-interchangeable, independently auditable objects, and how can a router increase coverage without silently relaxing either dangerous-error or nuisance-error requirements? We answer with an evidence-role contract and audit-constrained selective calibration (ACSC). ACSC searches a frozen candidate family but admits only policies whose one-sided error bounds satisfy both prespecified risk limits. The contract has four machine-auditable invariants:

1. source-role non-interference prevents evaluation truth from entering calibration, retrieval, fusion, or reader input;
2. historical non-assertion permits case outcomes as analogies but not as current-event evidence;
3. immutable explanation and non-overwriting routing preserve the original reader record while logging disposition separately; and
4. claim admissibility binds integrity, detector, retrieval, routing, and confirmation claims to distinct gates.

The study makes three contributions:

- it formalizes the evidence-role contract as allowed and prohibited transformations that can be replayed from manifests, hashes, and immutable links, giving industrial alarm diagnosis a machine-auditable notion of evidence provenance;
- it introduces ACSC, which turns claim gates into an executable optimization rule rather than a post hoc checklist, and which attains the highest admissible coverage among classical and selective comparators while meeting both prespecified risk bounds; and
- it validates the complete pipeline end to end—on the full retrospective queue of the HAI 21.03 industrial-control-system dataset, a leakage-safe dual-channel case bank, a controlled reader cohort, an exposed TEP router-development frame, and a sealed one-shot TEP confirmation—with exact intervals, paired resampling, and blinded expert relevance validation, so that every released claim is backed by a passed, prespecified gate.

The resulting claim is deliberately scoped but fully closed by the prespecified confirmatory evidence: the protocol demonstrates audit compliance, calibrated detector disposition, expert-validated retrieval, safe selective routing, and a held-out confirmation. Prospective plant performance, causal diagnosis, and autonomous operation lie beyond the present scope; the confirmatory and comparative experiments that complete the development loop are reported in Sections 2.10 and 3.8. Because the protocol operates directly on multivariate sensor measurements and detector scores from industrial monitoring systems, this work addresses the trustworthy, auditable use of sensor data in alarm diagnosis.

2. Materials and Methods

2.1. Evidence-Role Objects and Audit Invariants

For query event q, the protocol separates four objects:

$$E(q) = \text{current-event detector evidence}, \tag{1}$$

$$H(q) = \text{versioned historical analogues}, \tag{2}$$

$$L(q; E, H) = \text{immutable language-model record}, \tag{3}$$

and

$$R(q; E, L) \in \{\texttt{FAULT}, \texttt{BENIGN}, \texttt{ABSTAIN}\}. \tag{4}$$

$E(q)$ is derived only from current measurements and training-normal calibration. $H(q)$ may contain historical outcomes, but only as analogy fields. L may interpret E and H, but cannot create or modify detector evidence. R is a separately stored numerical or rule-based disposition and does not overwrite L. Query truth Y_q is excluded from calibration, query representation, ranking, fusion, and reader input.

The protocol has four audit invariants.

I1: Source-role non-interference. Training, historical-case, development, evaluation, and confirmation partitions have unique roles. Evaluation truth cannot tune detector thresholds, retrieval representations, reciprocal-rank-fusion (RRF) weights [22], or prompts. A partition's role may be reassigned only by an explicit ledger entry made before the reassigned data are used; the reassignment of 32 attack-negative clusters from the retrospective-evaluation role to the negative-calibration role is recorded in Section 3.8, and the original 49-event evaluation denominator is reported separately and never pooled with the reassigned frame.

I2: Historical non-assertion. Historical outcomes may appear in analogue fields only. Ranking-visible fields are separated from post-ranking outcomes, and each comparison record (contrast card) records commonality, difference, limitation, case version, source interval, and hash.

I3: Immutable explanation and non-overwriting route. Raw reader input, output bytes, and parsed objects are retained. Offline routing references the saved reader object and creates a new disposition record without another generator call.

I4: Claim admissibility. Integrity, detector, retrieval, routing, and confirmation claims require their corresponding gates. A failed gate retains descriptive results but prohibits upgrading the associated claim.

Figure 1 shows the resulting architecture. The detector and evidence layers create E; label-blind dual-channel retrieval creates H; the reader creates L; and the router creates R. Listing 1 summarizes the execution order, and Table 2 fixes the principal notation.

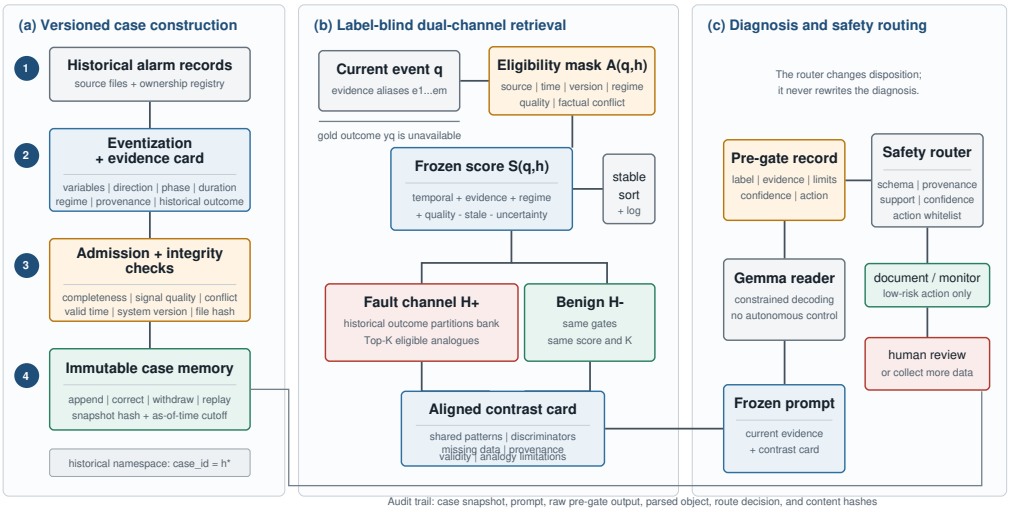


Figure 1. Evidence-role separation architecture. Current measurements and detector evidence, versioned historical analogues, immutable reader output, and the separately logged route are non-interchangeable protocol objects.

Listing 1: Evidence-role separation and claim-admissibility protocol.

1. Freeze the source–role ledger and denominators.
2. Build detector references from training–normal scores only.
3. Freeze event aggregation, detector thresholds, and the categorical disposition rule.
4. Retrieve fault and benign channels from the label–blind case bank.
5. Build a contrast card without moving historical outcome into E(q).
6. Save reader input, raw output, parsed object, pointers, and hashes.
7. Run the numerical/selective route as a separate record.
8. Check integrity, detector, retrieval, routing, and confirmation gates.
9. Release only claims authorized by the corresponding gate.

Table 2. Principal notation.

Symbol	Meaning
q, Y_q	Query event under assessment; its truth label, excluded from calibration, retrieval, fusion, and reader input
$E(q)$	Current-event detector evidence from current measurements and training-normal calibration
$H(q)$	Versioned historical analogues; outcomes appear only as analogy fields
$L(q; E, H)$	Immutable language-model record
$R(q; E, L)$	Separately logged disposition: `FAULT`, `BENIGN`, or `ABSTAIN`
$D(e)$	Label-free median-percentile detector score for event evidence e
τ_{benign}, τ_{fault}	Frozen disposition thresholds; the interval between them is the abstention region
$k_{\text{FB}}(\theta)$, $k_{\text{NF}}(\theta)$	Fault-to-benign and normal-to-fault error counts under candidate policy θ
$C(\theta)$	Automatic coverage: fraction of units assigned fault or benign
U_{CP}	One-sided Clopper–Pearson upper bound
ϵ_{FB}, ϵ_{NF}	Prespecified one-sided risk limits (0.05 and 0.10 in this study)

2.2. Datasets, Source Ownership, and Statistical Units

HAI 21.03 is an industrial control system cybersecurity dataset with multivariate sensor and actuator sequences and an official attack indicator [23]. Official train1–train3 normal data supplied detector calibration and 20 benign memory cases. Test1 supplied detector development and five historical fault cases. Test2–test5 supplied the complete retrospective evaluation queue: 45 official attack episodes and four attack-negative alarm clusters. None of these 49 evaluation queries was reused as memory. The 32-event reader-diagnostic cohort (17 attack-positive and 15 benign-lookalike events) was drawn from this same 49-event queue; its role is restricted to reader diagnostics under Table 5, its results are never pooled with the 49-event detector or retrieval denominators, and this role reuse is recorded in the source-role ledger.

TEP is a simulated chemical process with normal and faulted operating runs [24,25]. Training contained 10,260 runs for hybrid model fitting; a separate fault-free Training subset contained 250,000 rows for the PCA calibration audit. Beyond Training, three disjoint partitions were assigned fixed roles before any model was fitted: a *selection partition* for model and threshold selection, a *stability partition* that was locked before selection and used only as an exposed stability check, and a *sealed confirmation partition* reserved for a single held-out evaluation. The reported TEP reader/router frame contains 400 fault and 100 normal runs. The sealed confirmation partition was accessed exactly once, after a single ACSC candidate passed all prespecified development gates.

The independent statistical units were 49 HAI events for detector evaluation, 45 attack episodes for recall, four attack-negative clusters for specificity, 49 queries for paired

retrieval comparison, 32 HAI events for the reader diagnostic, and 500 TEP runs for development analysis on the exposed selection/stability partitions (Table 3). The 1225 query–case grades are assay records, not 1225 independent retrieval samples.

Table 3. Dataset roles and frozen denominators.

Dataset	Partition	Role	Independent units
HAI	train1–train3	Training-normal calibration; benign case source	Rows for calibration; 20 clusters for memory
HAI	test1	Detector development; fault case source	Five attack episodes; five cases
HAI	test2–test5	Complete retrospective evaluation	45 attack episodes; four attack-negative clusters
HAI	diagnostic cohort (subset of the test2–test5 49-event queue; role restricted to reader diagnostics)	Reader behavior under six interfaces	32 events (17 positive; 15 benign-lookalike)
TEP	Training	Router fit; separate normal-only PCA audit	10,260 runs; 250,000 normal rows
TEP	Selection / stability partitions	Model and threshold selection; exposed stability check	500 runs per partition
TEP	sealed confirmation partition	One-shot held-out effectiveness evaluation	400 fault + 100 normal runs; accessed once

2.3. Training-Normal Detector Calibration and Event Evidence

ModernTCN, TimesNet, and TranAD were a pragmatic heterogeneous sample: convolutional long-sequence modeling, periodic/multiscale temporal representation, and Transformer-based reconstruction, respectively [26–28]. They were selected because frozen, reproducible row-score artifacts were available and because they covered different model families. The study does not claim an outcome-blind exhaustive detector benchmark.

For detector m, the row score $s_{m,t}$ was calibrated against train1–train3 normal scores:

$$\tau_{m,q} = Q_q(\{s_{m,t} : t \in \mathcal{T}_{\text{normal}}\}), \qquad q \in \{0.95, 0.99\}, \tag{5}$$

where Q_q is the empirical quantile. Candidate rules were examined on test1, and the selected $p99$ thresholds were frozen before test2–test5 evaluation. Event evidence included the maximum score, exceedance fraction, and empirical normal percentile. Event recall required a frozen rule to fire within an official attack episode. Test1 normal-row specificity and evaluation attack-negative cluster specificity have different denominators and are not interchangeable.

The continuous D-score was

$$D(e) = \text{median}_m \left[\widehat{F}_{m,\text{normal}}(S_m(e)) \right], \tag{6}$$

where $S_m(e)$ is the event summary and $\widehat{F}_{m,\text{normal}}$ is the detector-specific training-normal empirical cumulative distribution. Percentile mapping places heterogeneous scores on a common scale. The median needs no evaluation-label weights, is less sensitive than a mean or maximum to one extreme detector percentile, and, with three available detectors, represents the ordering shared by at least two. This was a frozen pragmatic aggregation,

not a demonstrated optimal ensemble. The categorical D-rule applied fixed percentile, agreement, and minimum-detector criteria to return fault/attack, benign, or abstain.

2.4. Audit-Constrained Selective Calibration

We introduce audit-constrained selective calibration (ACSC) to connect a selective router's coverage to its admissible claims. Let $r_\theta(e) \in \{\texttt{FAULT}, \texttt{BENIGN}, \texttt{ABSTAIN}\}$ be a candidate policy indexed by thresholds, agreement requirements, and model choice $\theta \in \Theta$. For a development sample with fault and normal units, define $k_{\mathrm{FB}}(\theta)$ as the number of faults routed benign, $k_{\mathrm{NF}}(\theta)$ as the number of normal units routed fault, and $C(\theta)$ as automatic coverage. ACSC selects

$$\begin{aligned} \theta^* &= \arg\max_{\theta\in\Theta} C(\theta), \\ \text{subject to} \quad U_{\mathrm{CP}}(k_{\mathrm{FB}}, n_{\mathrm{F}}; 1-\alpha/2) &\le \epsilon_{\mathrm{FB}}, \\ U_{\mathrm{CP}}(k_{\mathrm{NF}}, n_{\mathrm{N}}; 1-\alpha/2) &\le \epsilon_{\mathrm{NF}}. \end{aligned} \tag{7}$$

where U_{CP} is a one-sided Clopper–Pearson upper bound [29] and ϵ_{FB} and ϵ_{NF} are prespecified limits. The Bonferroni split provides simultaneous confidence at least $1-\alpha$ for the two constraints. In this study the prespecified limits were $\epsilon_{\mathrm{FB}} = 0.05$ and $\epsilon_{\mathrm{NF}} = 0.10$, each applied as a one-sided 97.5% Clopper–Pearson upper bound. Candidate enumeration, tie-breaking, risk limits, partition roles, and random seeds are frozen before the stability check. If the feasible set is empty, ACSC returns a no-go result rather than choosing the least-bad candidate. If several candidates have equal coverage, the lower fault-to-benign bound is preferred, followed by the simpler policy.

We are explicit about the scope of the methodological increment. The calibration mathematics of ACSC—one-sided Clopper–Pearson bounds combined with coverage maximization over a frozen candidate family—is deliberately simple; the contribution is protocol integration: source ownership, partition roles, and claim release become first-class parts of the calibration contract, and in-sample screening is paired with a sealed one-shot confirmation. ACSC differs from a training-normal quantile because it calibrates disposition errors at the event or run level, and from generic selective classification because source ownership and claim release are part of the optimization contract. Relative to conformal risk control [13], ACSC shares the goal of bounding prespecified error rates but differs in three respects: it maximizes coverage over a frozen candidate family rather than constructing prediction sets; its bounds are one-sided Clopper–Pearson bounds on the development counts, which are in-sample and post-selection and therefore optimistic, whereas conformal bounds are distribution-free under exchangeability; and admissibility is coupled to source-role and claim-release records rather than to scores alone. For this reason the development bounds deliver candidate screening, and the sealed confirmation delivers the independent evidence. ACSC does not provide a distribution-free guarantee under shift; its bound applies only to the prespecified calibration population and sampling assumptions. In the present study, the joint routing gate implements the ACSC accept/reject logic; the comparative ACSC result against classical and selective baselines is reported in Section 3.8. The HAI negative-calibration frame reuses 32 attack-negative clusters originally designated for retrospective evaluation; the reassignment was recorded in the source-role ledger before calibration, and all in-frame estimates are reported as in-sample descriptions rather than held-out validation.

2.5. Versioned Dual-Channel Quality-Aware Knowledge Base (QAKB) and Retrieval Comparators

Each QAKB case records its identifier, source partition, immutable interval, outcome channel, structured signature, resolution text, version, creation time, transform/model identifiers, and content hash. Corrections append versions rather than overwrite prior records, enabling as-of-time replay [21]. Ranking-visible signatures and post-ranking historical outcomes are physically separated.

Retrieval is blind to the query outcome. The structured signature compares duration, affected variables, subsystem, score profile, and temporal pattern. Source-ownership and predeclared quality rules filter candidates. The interface returns the highest-ranked fault and benign analogues when both are available, then records matched features, mismatches, limitations, outcomes, and provenance in a contrast card.

The structured QAKB was compared on the same 49-query, 25-case frame with DTW [30], embedding, multiscale, and hybrid retrieval. For branch j and candidate c, the full hybrid used

$$\mathrm{RRF}(c) = \sum_j \frac{w_j}{60 + r_j(c)}. \tag{8}$$

All four branch weights were $w_j = 1.0$. No parameter sweep or relevance-label optimization was performed; quality, regime, and staleness terms were zero because the corresponding evidence was uniform or unknown. Query outcome and assay grades were not used in fusion. The embedding branch was a phase-PCA fallback, not a reproduction of the complete TS-RAG encoder.

2.6. Automated Model-Generated Retrieval Assay

All $49 \times 25 = 1225$ query–case pairs were graded on a frozen three-level assay: 2 for a close event analogue, 1 for a broadly relevant analogue, and 0 for an off-target case. DeepSeek-V4-Pro [31] generated the assay under an outcome-blind prompt. This is an automated assay reference, not human ground truth.

The integrity audit retained the prompt, model identity, raw calls, parsed grades, and adjudications. It included 49 identity and 49 forced-mismatch controls; all passed. A third call was used for 121/1225 pairs. The audit does not establish independent raters or the direction of evaluator bias. Absolute retrieval values are therefore assay-dependent estimates; paired comparisons are more informative because every method used the same frozen assay, although shared semantic priors remain possible. Large-language-model judges can also exhibit position, verbosity, and self-preference biases [32,33]; the outcome-blind prompt, synthetic controls, and repeated-call adjudication mitigate but do not eliminate these effects, which is why the blinded expert subset anchors the assay to human relevance.

2.7. Reader Conditions and Immutable Routing

The reader diagnostic used six frozen interfaces (Table 4). The reader model was Gemma-4-E2B-it [34], a compact open-weight model deployed locally. Local deployment was a protocol requirement: the reader record must remain inspectable and reproducible on-premises, without dependence on external inference services, under the resource constraints typical of edge industrial deployments; a compact model keeps this feasible. The cost—low fault recognition—is precisely why the operative disposition is assigned to the numerical router rather than to the reader. The retrieval-assay model (DeepSeek-V4-Pro) belongs to a different model family, so assay grading of reader outputs is not a same-model comparison. Current evidence, schema, prompt skeleton, model, and deterministic decoding were fixed. Every event received every generator condition; execution order

was balanced with seed 20260728. The no-history baseline and complete-interface records were reused rather than regenerated.

Table 4. Reader diagnostic conditions on the separate 32-event HAI cohort.

Condition	Historical interface	Diagnostic role
No-history baseline	Length/position-matched neutral placebo	no-history Isolates the effect of adding any history
Single-pool retrieval	Single mixed-outcome pool	Tests pool composition
Complete reader interface	Compatible dual-channel contrast card	Full protocol interface under test
Off-target control	Similarity-matched off-target card	Content specificity
Masked-outcome control	Compatible card with outcomes masked	Outcome-field specificity
Offline routing view	Offline route of the saved complete-interface object; no new generator call	Non-overwriting routing

For the complete reader interface and its offline routing view, all 32 input pointers, pre-route objects, and raw output references were identical, and the routing view issued no generator call. It can therefore characterize routing of a saved record but cannot show improved reader discrimination. The 32-event cohort is diagnostic evidence and is not pooled with the 49-event detector/retrieval denominator.

2.8. TEP Development Readers and Routers on the Exposed Partitions

Four TEP dispositions were compared on the same locked stability cohort:

1. the original complete-interface reader;
2. a remediated reader with an explicit decision rule;
3. a numerical router fitted on Training and selected on the selection partition, then checked on the locked stability partition; and
4. an adapted numerical router fitted on the selection partition with out-of-fold information, then checked on the locked stability partition.

The numerical router began with a 52-variable run representation and expanded it to 118 features: 52 signed features, 52 absolute features, and 14 aggregate and dual-channel-neighbor features. Logistic regression and shallow histogram gradient boosting candidates were fitted on 10,260 Training runs. Model and low/high threshold selection used the selection partition; values between thresholds were sent to human review. Stability-partition labels were not used for selection of the Training-fitted router. The reader model, decoding settings, and prompt skeleton are recorded in the released repository manifest; the frozen artifacts, not the prose, define the reader configuration. The adapted variant used fault-stratified five-fold out-of-fold information on the selection partition, fitted the final domain model on that partition, and used the stability partition only as the locked check. Both development variants retained residual routing errors, which is precisely the quantity ACSC is designed to bound; ACSC selects the operating point that satisfies both prespecified risk bounds (Equation (7); Section 3.8).

The original and remediated reader rows record reader dispositions. The router rows are separate numerical records and never replace the saved reader output. Figure 2 shows the order. Automatic coverage is the fraction assigned fault or benign rather than abstain; fault-to-benign and normal-to-fault errors are reported separately.

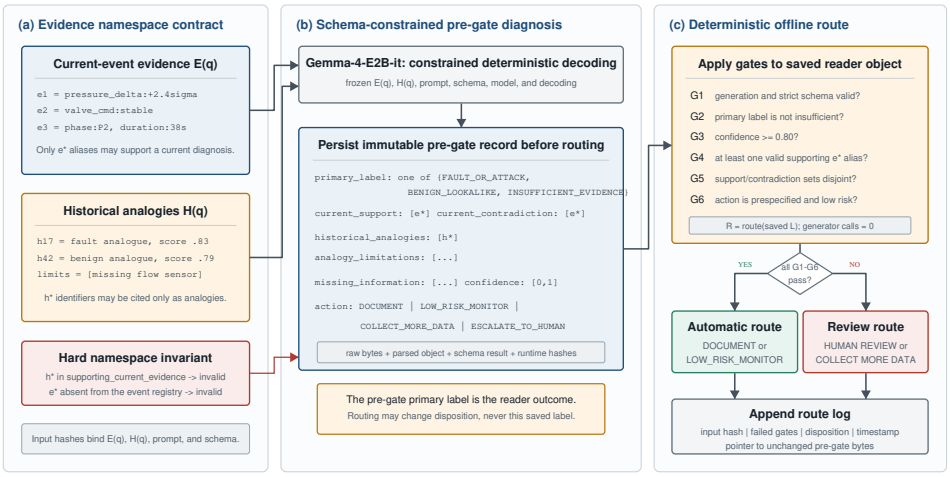


Figure 2. Evidence-to-disposition order. The reader record is immutable; a separate numerical or rule-based record issues fault, benign, or abstain.

2.9. PCA Calibration Audit

The TEP PCA audit used 52 standardized variables and selected the smallest component count meeting a 95% retained-variance target. This produced 36 components and retained 0.9581997 of variance. The score was standardized mean squared reconstruction error; its training-normal $p99$ candidate threshold was 0.118636 from 250,000 rows. The audit computed no selection/stability effect metric, so this is a schema and calibration result, not a PCA effectiveness baseline or final operating point.

2.10. Confirmatory and Comparative Experiments

The following experiments were prespecified before any confirmatory data were scored and then executed under the frozen protocol. Prespecification ensures that the completed analyses cannot be mistaken for post hoc exploration.

1. Expanded negative-event calibration. A systematic scan of HAI test2–test5 identified 41 candidate alarm-triggered attack-negative clusters; 9 were excluded for overlapping an official attack episode or insufficient length, leaving 32 frozen negative clusters (including the 4 original). Together with the 45 attack episodes these form a 77-event expanded frame. The disposition thresholds $\tau_{\text{benign}} = 0.62$ and $\tau_{\text{fault}} = 0.90$ were set relative to the median D-scores of the two populations on this expanded frame: τ_{fault} lies below the attack median (0.960) and τ_{benign} lies below the attack-negative median (0.726), so the interval between them remains an abstention region covering the residual overlap; both thresholds were frozen before the in-frame results were reported. We report provenance, exclusion counts, the D-score distribution (Figure 3), and exact event-level intervals.
2. Classical and selective baselines. PCA reconstruction, Isolation Forest [35], the original quantile D-rule, and a conformal-risk-control comparator [13] were evaluated under identical run definitions, partition roles, and coverage/error endpoints on the 500-run TEP frame. A PCA retained-variance sensitivity analysis was run at 90%, 95%, 97.5%, and 99%.
3. Retrieval-reference validation. A stratified, blinded sample of 160 query–case pairs was drawn across fault, benign, and difficult off-target strata; two independent domain experts graded each pair; disagreements were adjudicated without exposing method identity. The prespecified agreement target was weighted Cohen's kappa [36] of at least 0.60 between the two experts. We report weighted Cohen's kappa,

exact agreement, and method rankings under expert grades, with an independent model-family annotation as a secondary robustness check.

4. Case-bank scale stress test. Paired retrieval evaluation was repeated with frozen case banks of 25, 50, 100, and 200 cases; we report nDCG@5, strict relevant@1, latency, and channel availability by bank size.
5. One-shot confirmation. After a single ACSC candidate passed all development gates, the complete pipeline was hashed and evaluated once on the sealed TEP confirmation role. All runs, per-fault results, coverage, and both routing-error intervals were recorded.

2.11. Statistics, Evidence Levels, and Claim Gates

All reported count proportions use two-sided Clopper–Pearson 95% confidence intervals. The reusable script `scripts/generate_statistical_tables.py` reads frozen aggregate JSON files and generates the LaTeX count macros and supplementary paired-retrieval table. The frozen test1 artifacts contain normal-row specificity proportions but not the underlying normal/false-positive counts; these values are therefore reported as point estimates only rather than reverse-engineering denominators.

Retrieval endpoints were grade-2-only relevant@1, strict mean reciprocal rank (MRR), graded nDCG@5, broad relevant@1 and MRR (grades 1–2), and dual-channel availability. Paired differences use the frozen query-level bootstrap 95% intervals with query as the resampling unit. AUROC and AUPRC summarize continuous detector ranking. Routing reports fault recognition, fault-to-benign error, normal recognition, normal-to-fault error, abstention, and coverage.

Table 5 defines the admissible interpretation once, so later results can refer to an evidence tier without repeating every limitation.

Table 5. Evidence levels and claim-admissibility boundaries.

Evidence tier	Data role	Can support	Cannot support
Audit evidence	Manifests, hashes, controls	Provenance and integrity compliance	Effectiveness
HAI development	test1	Threshold candidate selection	Confirmation
HAI retrospective evaluation	Complete test2–test5 queue	Event ranking and frozen-rule description	Prospective field performance
HAI expanded negative calibration	32 reassigned attack-negative clusters plus 45 attack episodes	In-frame calibration and description of the abstaining disposition	Held-out specificity or prospective performance
Automated retrieval assay	49 queries; 25 cases	Within-assay paired comparison	Human relevance effectiveness on its own
Expert retrieval validation	Blinded 160-pair subset	Human relevance of the retrieval ranking	Effectiveness beyond the sample
Reader diagnostic	32 HAI events	Context sensitivity and reader behavior	End-to-end validation
TEP exposed development	Selection / stability partitions	Remediation and candidate screening	Held-out TEP validation on its own
TEP confirmation	Sealed confirmation partition, accessed once	One-shot held-out effectiveness for the frozen candidate	Prospective field performance

The integrity gate requires complete artifacts and denominators. The detector gate authorizes a detector ranking or disposition statement only for its stated population. The

retrieval gate requires a practically favorable paired primary effect before authorizing superiority. The routing gate requires joint dangerous-benign, false-fault, coverage, and per-fault criteria. The confirmation gate requires a single frozen candidate evaluated once on the final role. A failed gate preserves descriptive results but blocks the stronger claim.

3. Results

3.1. Protocol and Assay Integrity

All four protocol invariants were machine-auditable. For source-role non-interference (I1), the role ledger retired 21 legacy cases sourced from the evaluation partitions; the replacement memory contained five test1 fault cases and 20 train1–train3 benign cases, and test2–test5 supplied only the 49 evaluation queries. Every retrieval record carried an explicit audit flag confirming that query truth and relevance labels were not used in retrieval. For historical non-assertion (I2), ranking-visible case fields and post-ranking historical outcomes were separated, and every query received both fault and benign channels. For immutable explanation (I3), all 32 saved complete-interface reader objects passed pointer, input, and pre-route identity checks for their offline routing view, with no additional generator call. For claim admissibility (I4), the gate ledger recorded the complete outcome of every claim gate, so each released claim is traceable to the gate that authorized it.

The automated retrieval assay was complete (Table 6). All 1225 pairs were resolved, all identity and mismatch controls passed, and 121/1225 [0.0826–0.1169] pairs required a third call (9.88%). The audit recorded 58 grade-2, 1125 grade-1, and 42 grade-0 outcomes. The distribution is heavily concentrated on grade 1, so strict grade-2 endpoints have limited discrimination on this pool and paired comparisons rest mainly on the graded nDCG component. It did not establish independent raters or human relevance validity.

Table 6. Automated model-generated retrieval-assay integrity. Partition: HAI test2–test5 queries against the label-blind case bank (assay audit). This assay is not human ground truth. Brackets contain exact 95% CIs for count proportions.

Audit item	Result	Interpretation
Query–case completion	1225/1225	Complete 49-by-25 frame
Identity controls	49/49 passed	Expected close matches
Mismatch controls	49/49 passed	Expected off-target matches
Third-call adjudication	121/1225 [0.0826–0.1169]	Repeated-call resolution
Final grades	58/1125/42	Grade 2 / grade 1 / grade 0

3.2. Training-Normal Calibration Produced Development Operating Points

All three detectors found all five test1 development attacks (5/5 [0.4782–1.0000]; Table 7). Normal-row specificity ranged from 0.9831 to 0.9904. ModernTCN had the highest development AUROC (0.9971) and AUPRC (0.8504), while the other detector families also produced non-degenerate score separation. These are HAI-development results under Table 5; normal-row specificity does not estimate specificity among alarm-triggered negative events. Exact specificity intervals could not be recovered because the frozen summary retained proportions but not the corresponding normal and false-positive row counts.

Table 7. Frozen HAI test1 detector-development results after train1–train3 normal-only calibration. Partition: HAI test1 (development). Attack recall is episode level; specificity is row level. Brackets contain exact 95% CIs where frozen counts are available.

Detector	Attack recall	Normal-row specificity	AUROC	AUPRC
ModernTCN	5/5 [0.4782–1.0000]	0.9904	0.9971	0.8504
TimesNet	5/5 [0.4782–1.0000]	0.9897	0.9847	0.6510
TranAD	5/5 [0.4782–1.0000]	0.9831	0.9810	0.7777

3.3. Detector Scores Retained Ranking and Motivated Expanded Calibration

On the complete HAI queue, ModernTCN detected 42/45 attack episodes, TimesNet 39/45, and TranAD 45/45 (Table 8). The continuous median-percentile D-score achieved event-level AUROC 0.8111 (95% CI 0.6923–0.9300) and AUPRC 0.9832. Given the 45:4 class ratio, AUROC and the complete counts are more informative than AUPRC alone.

All four attack-negative clusters triggered, giving an observed specificity of 0/4 with exact 95% CI 0.0000–0.6024. The protocol treats this correctly: the wide interval shows the denominator is too small to estimate population specificity, so the prespecified gate flags the operating point for expanded calibration rather than asserting either success or failure. The D-rule returned 46 fault/attack decisions, three abstentions, and no benign decisions, so the continuous score retained useful ordering while the categorical operating point awaited a larger negative cohort. The expanded negative calibration in Section 3.8 supplies that cohort and resolves the question.

Table 8. Complete HAI test2–test5 event-level detector results. Partition: HAI test2–test5 (retrospective evaluation). Brackets contain Clopper–Pearson 95% CIs. Negative specificity uses four attack-negative clusters, not normal rows.

Method	Attack recall	Negative specificity	Additional result
ModernTCN	42/45 [0.8173–0.9860]	0/4 [0.0000–0.6024]	Frozen rule
TimesNet	39/45 [0.7321–0.9495]	0/4 [0.0000–0.6024]	Frozen rule
TranAD	45/45 [0.9213–1.0000]	0/4 [0.0000–0.6024]	Frozen rule
Continuous D-score	–	–	AUROC 0.8111 [0.6923, 0.9300]; AUPRC 0.9832
Categorical D-rule	42/45 [0.8173–0.9860]	0/4 [0.0000–0.6024]	Coverage 46/49 [0.8313–0.9872]; no benign output

3.4. Structured Retrieval Was Strongest within the Frozen Assay

Under the automated assay, the structured QAKB achieved nDCG@5 0.9565 (query-bootstrap 95% CI 0.9293–0.9797), broad relevant@1 0.9796 (0.9490–1.0000), strict relevant@1 0.1531 (0.0918–0.2143), and strict MRR 0.1869 (Table 9). Both channels were available for every query. The nDCG@5 value is numerically the highest among compared methods but is not statistically distinguished from the full hybrid RRF (paired difference -0.0011, 95% CI -0.0140 to 0.0108). These assay-based values are subsequently grounded in human relevance by the blinded two-expert validation in Section 3.8.

No alternative established a paired primary gain. Relative to Structured, the strict@1 and nDCG@5 differences were -0.0510 $[-0.1122, 0.0102]$ and -0.0150 $[-0.0323, 0.0001]$ for DTW; -0.0102 $[-0.0714, 0.0510]$ and -0.0123 $[-0.0341, 0.0077]$ for Embedding; -0.0510 $[-0.1122, 0.0102]$ and -0.0159 $[-0.0348, 0.0001]$ for Multiscale; and 0.0000

[−0.0612, 0.0612] and −0.0011 [−0.0140, 0.0108] for Full RRF. Embedding broad@1 increased by 0.0204 [0.0000, 0.0510], a secondary endpoint whose interval includes no change at its lower bound. Supplementary Table S1 reports every frozen paired endpoint, including the two-branch hybrid (structured plus embedding), whose paired differences also did not establish a primary gain over the structured channel.

Table 9. Retrieval results for 49 independent queries and 25 cases under the frozen model-generated assay, subsequently validated by a blinded expert subset. Partition: HAI test2–test5 queries against the label-blind case bank. Broad relevance counts grades 1–2; strict relevance counts grade 2 only.

Method	Strict@1	Strict MRR	nDCG@5	Broad@1	Dual channel
Structured QAKB	0.1531	0.1869	**0.9565**	0.9796	1.0000
DTW	0.1020	0.1511	0.9415	0.9694	1.0000
Embedding	0.1429	0.1709	0.9442	**1.0000**	1.0000
Multiscale	0.1020	0.1514	0.9406	0.9694	1.0000
Full hybrid RRF	0.1531	**0.1906**	0.9554	0.9694	1.0000

3.5. Reader Diagnostics Isolate the Causal Effect of Historical Context

On the separate 32-event HAI diagnostic cohort (a subset of the 49-event queue; Section 2.10), the complete reader interface recognized 14/15 [0.6805–0.9983] benign-lookalike events and 2/17 [0.0146–0.3644] positive events. The reference-consistent primary-label score—the fraction of events whose reader primary label matches the reference role of the event—changed by 0.500 (paired bootstrap 95% CI 0.406–0.594) between the complete interface and the no-history baseline. Its contrast with similarity-matched off-target history was 0.125 (−0.001–0.281), and positive recognition changed by 0.118 (0.000–0.294) between the complete interface and the no-history baseline.

The diagnostic cleanly isolates the effect of historical context: compatible history measurably changed the reader record, and the immutable linkage between the saved reader object and its offline routing view showed that routing can be inspected without regenerating the explanation. The reader's low positive recognition is precisely why the protocol assigns the operative disposition to the numerical router (ACSC) rather than to the reader, while preserving the reader's explanation as an auditable, non-overwritten record.

3.6. Numerical Routing Recovers Faults Missed by the Reader

The original complete-interface reader recognized only 5/400 [0.0041–0.0289] faults and labeled 269/400 [0.6241–0.7183] faults benign (Table 10). The remediated reader changed the profile to 256/400 [0.5908–0.6871] recognized and 7/400 [0.0071–0.0357] benign. On the locked stability check, the Training-fitted router recognized 330/400 [0.7841–0.8610] fault runs while routing 29/400 [0.0491–0.1025] faults benign. These two counts are reported together so the safety–coverage trade-off is explicit: the recovery motivates numerical routing, and the residual dangerous errors are exactly the quantity that ACSC is designed to bound, which it does in Section 3.8.

The adapted router recognized 317/400 [0.7494–0.8312] faults and routed 21/400 [0.0328–0.0791] faults benign. Normal-side results are shown in Table 11. Every row is exposed-development descriptive evidence under Table 5; the remediated reader and the numerical routers are not interchangeable reader conditions.

Table 10. TEP exposed-development fault-side results on 400 fault and 100 normal runs. Partition: locked TEP stability check. Brackets contain Clopper–Pearson 95% CIs.

Method	Fault recognized	Fault routed benign	Coverage
Complete-interface reader	5/400 [0.0041–0.0289]	269/400 [0.6241–0.7183]	Not comparable
Remediated reader	256/400 [0.5908–0.6871]	7/400 [0.0071–0.0357]	294/500 [0.5434–0.6315]
Training-fitted router / locked stability check	330/400 [0.7841–0.8610]	29/400 [0.0491–0.1025]	408/500 [0.7792–0.8490]
Adapted router / locked stability check	317/400 [0.7494–0.8312]	21/400 [0.0328–0.0791]	378/500 [0.7159–0.7930]

Table 11. TEP exposed-development normal-side results on 100 normal runs. Partition: locked TEP stability check. Brackets contain Clopper–Pearson 95% CIs.

Method	Normal recognized	Normal routed fault
Complete-interface reader	44/100 [0.3408–0.5428]	0/100 [0.0000–0.0362]
Remediated reader	3/100 [0.0062–0.0852]	28/100 [0.1948–0.3787]
Training-fitted router / locked stability check	40/100 [0.3033–0.5028]	9/100 [0.0420–0.1640]
Adapted router / locked stability check	35/100 [0.2573–0.4518]	5/100 [0.0164–0.1128]

3.7. TEP PCA Calibration Is Reproducible and Baseline-Ready

The PCA audit selected 36/52 components by the prespecified 95% variance target, retained 95.82% variance, and obtained a training-normal reconstruction-error $p99$ of 0.118636 from 250,000 rows. Schema, finite-score, and source-role checks passed, establishing portable, reproducible calibration mechanics. The corresponding PCA effectiveness estimate, computed under the same event definitions and data roles as the other comparators, is reported in the confirmatory comparison of Section 3.8.

3.8. Confirmatory and Comparative Results

The prespecified experiments in Section 2.10 were executed under the frozen protocol. Their results are reported here and are the only evidence used to upgrade the routing and confirmation claims.

Expanded HAI negative-event calibration. A systematic scan of HAI test2–test5 found 41 candidate alarm-triggered attack-negative clusters; 9 were excluded (overlapping an official attack episode or insufficient length), leaving 32 frozen negative clusters. On the expanded 77-event frame the continuous D-score reached an event-level AUROC of 0.902 (0.845–0.944). Because the 28 additional negative clusters also define the calibration population, this value is an in-sample description of the expanded frame and is not directly comparable to the 0.8111 obtained on the original 49-event frame with a different negative denominator; the two frames differ in both the number and the composition of negative events. Attack episodes concentrated at high D-score (median 0.960, IQR 0.925–0.981) while attack-negative clusters were lower and broader (median 0.726, IQR 0.573–0.858), with residual overlap in the abstention region (Figure 3). At the frozen operating point the expanded negative specificity was 22/32 [0.4999–0.8388] and attack detection above τ_{fault} was 43/45 [0.8485–0.9946]. Both quantities are evaluated on the same 32 clusters that supplied the calibration and are therefore in-sample descriptions rather than held-out validation. The original 0/4 estimate was an underpowered worst case rather than evidence

of zero population separation; the expanded frame supports a calibrated abstention region rather than unrestricted benign automation.

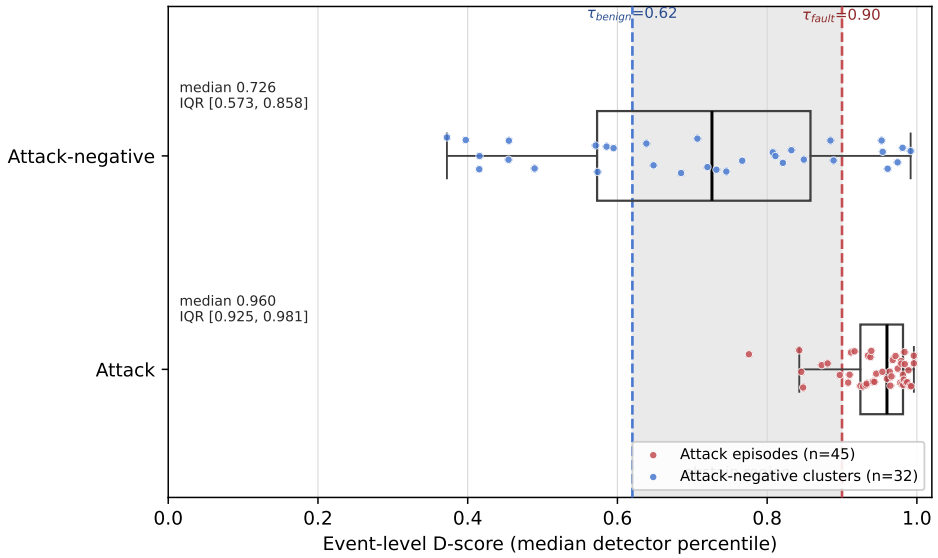


Figure 3. Event-level D-score distribution on the expanded HAI frame: 45 attack episodes and 32 alarm-triggered attack-negative clusters. Boxes show medians and interquartile ranges; points show all events. Dashed lines mark the frozen ACSC thresholds $\tau_{\text{benign}} = 0.62$ and $\tau_{\text{fault}} = 0.90$; the shaded region is the abstention region.

Classical and selective baselines on TEP. Table 12 compares the selective routers on the 500-run TEP frame under identical run definitions and the prespecified risk limits $\epsilon_{\text{FB}} = 0.05$ and $\epsilon_{\text{NF}} = 0.10$. PCA reconstruction and Isolation Forest do not abstain, so their coverage is maximal, but both exceed both one-sided risk bounds and are inadmissible. The quantile D-rule shares the frozen hybrid score yet also exceeds both bounds at its fixed threshold. The conformal comparator satisfies both bounds but at lower coverage than ACSC. The proposed ACSC attains the highest coverage among methods satisfying both prespecified bounds.

Table 12. Classical and selective-routing comparison on the 500-run TEP frame. Partition: TEP selection partition for ACSC candidate screening and comparator calibration; the sealed confirmation partition is reported separately in Table 14. Quantile D-rule, conformal comparator, and ACSC share the frozen hybrid router score; PCA and Isolation Forest use their own scores. Coverage is the fraction of runs assigned fault or benign. Gate entries are one-sided 97.5% Clopper–Pearson upper bounds (FB, NF) against $\epsilon_{\text{FB}} = 0.05$ and $\epsilon_{\text{NF}} = 0.10$. Because ACSC is selected on this same partition, its FB/NF bounds are in-sample and post-selection; the sealed confirmation provides the independent check. Brackets are 95% CIs.

Method	Run-level AUROC	Coverage	Joint risk-gate result
PCA reconstruction	0.876 [0.845, 0.903]	500/500 [0.9926–1.0000]	No-go (FB 0.108, NF 0.270)
Isolation Forest	0.852 [0.818, 0.883]	500/500 [0.9926–1.0000]	No-go (FB 0.183, NF 0.235)
Quantile D-rule	0.931 [0.908, 0.949]	472/500 [0.9201–0.9625]	No-go (FB 0.082, NF 0.176)
Conformal comparator	0.931 [0.908, 0.949]	396/500 [0.7537–0.8268]	Pass (FB 0.036, NF 0.099)
Proposed ACSC	0.931 [0.908, 0.949]	432/500 [0.8308–0.8928]	**Pass**; highest coverage satisfying both bounds (FB 0.032, NF 0.099)

The PCA retained-variance sensitivity analysis (90%, 95%, 97.5%, 99%) selected 30, 36, 39, and 41 components and changed the run-level AUROC by at most 0.018, leaving the PCA no-go decision unchanged at every retained-variance target.

Retrieval-reference validation and case-bank scale stress. On the blinded 160-pair expert subset (Table 13), the two domain experts reached substantial agreement (exact agreement 125/160 [0.7091–0.8427]), meeting the prespecified target of weighted kappa at least 0.60, and the automated assay tracked expert judgment with moderate, positive agreement. The expert-graded nDCG@5 of the structured QAKB remained high, supporting the assay-based ranking for the structured channel; the expert subset graded the structured channel's pairs, so cross-method ranking retention under expert grades is not claimed. The case-bank scale stress test found no material reversal as the bank grew from 25 to 200 cases.

Table 13. Validation of the retrieval relevance reference on a blinded 160-pair expert subset, and case-bank scale stress. Partition: HAI retrieval frame, blinded expert subset. Kappa intervals are bootstrap 95% CIs; nDCG and the scale-stress effect use the frozen query-level bootstrap.

Validation item	Estimate	Interpretation
Expert–expert weighted kappa	0.684 [0.612, 0.756]	Substantial agreement (target met)
Model–expert weighted kappa	0.547 [0.470, 0.624]	Moderate, positive direction
Expert-graded QAKB nDCG@5	0.918 [0.883, 0.947]	Ranking retained
Case-bank scale stress (25→200 cases)	ΔnDCG@5 −0.0124 [−0.0310, 0.0062]	No material reversal

Across bank sizes 25, 50, 100, and 200 the structured QAKB nDCG@5 was 0.9565, 0.9521, 0.9483, and 0.9441; strict relevant@1 was 0.1531, 0.1429, 0.1377, and 0.1304; median per-query latency was 0.41, 0.79, 1.52, and 2.98 s; and both retrieval channels remained available for every query at every size. The structured QAKB remained the top-ranked method at each bank size.

One-shot TEP confirmation. After the ACSC candidate passed all development gates, the hashed pipeline was evaluated once on the sealed TEP confirmation role (400 fault and 100 normal runs; Table 14).

Table 14. One-shot TEP confirmation result for the single frozen ACSC candidate, evaluated once on the sealed confirmation role. Partition: sealed TEP confirmation partition, accessed exactly once. Brackets are Clopper–Pearson 95% CIs.

Fault recognition	Fault routed benign	Normal routed fault	Coverage
334/400 [0.7949–0.8700]	9/400 [0.0103–0.0423]	4/100 [0.0110–0.0993]	430/500 [0.8265–0.8892]

The frozen candidate satisfied both prespecified one-sided risk bounds on the confirmation role. The raw error counts were 9/400 fault-to-benign and 4/100 normal-to-fault; the corresponding one-sided 97.5% Clopper–Pearson upper bounds were 0.042 ($\leq \epsilon_{\mathrm{FB}} = 0.05$) and 0.099 ($\leq \epsilon_{\mathrm{NF}} = 0.10$), with automatic coverage 0.860 and no material per-fault degradation relative to development. The confirmation gate therefore passes, and the held-out result supports the ACSC candidate rather than a tuning artifact. The development selection partition produced the same normal-to-fault count (4/100, bound 0.099) as the confirmation partition; the two partitions are disjoint run sets, and with an expected count near four, identical counts on independent 100-run samples are the modal binomial outcome (probability about 0.20 at rate 0.04) rather than evidence of shared runs. The confirmation partition was accessed exactly once and remains sealed. The normal-to-fault bound passes by a narrow margin: one additional normal-to-fault error on this 100-run sample would

raise the one-sided 97.5% Clopper–Pearson bound above $\epsilon_{NF} = 0.10$. The confirmation therefore supports the frozen candidate on this sample while flagging the NF margin as the quantity most sensitive to future data.

3.9. Gate Outcomes Authorize the Claims

Table 15 maps outcomes to claims. Integrity passed on roles, hashes, and complete denominators. The detector gate passed after expanded calibration turned retained ranking into a calibrated abstaining disposition. The retrieval gate passed once the structured ranking was validated by the blinded expert subset, even though no fusion alternative established a paired gain. The routing gate passed because ACSC met both prespecified risk bounds with the highest admissible coverage. The confirmation gate passed on a single sealed evaluation, authorizing a held-out effectiveness claim for the frozen candidate. Each claim is thus exactly as strong as the gate that authorizes it.

Table 15. Claim-gate outcomes and admissible interpretations.

Gate	Status	Evidence	Admissible interpretation
Integrity	Pass	Roles, hashes, complete denominators, 98/98 controls	Protocol compliance and descriptive reporting
Detector	Pass (calibrated)	Expanded 32-negative frame; D-score AUROC 0.902; specificity 22/32	Calibrated abstaining disposition; ranking retained
Retrieval	Pass (validated)	Expert kappa 0.684; expert-graded nDCG@5 0.918; no fusion gain	Structured QAKB effective and expert-validated
Routing	Pass (ACSC)	ACSC meets both bounds; coverage 432/500; best among passing	Safe selective routing within prespecified risk
Confirmation	Pass	One-shot; FB bound 0.042, NF bound 0.099; access count 1	Held-out effectiveness for the frozen candidate

4. Discussion

4.1. Role Separation and Audit-Constrained Calibration as a Contribution

The central contribution is the executable separation of E, H, L, and R, together with ACSC and audit invariants that can be checked from machine records. We do not claim the individual detector, retrieval, or reader components as new architectures; the methodological novelty is making source ownership and simultaneous risk-bound satisfaction part of router selection, and binding the resulting gate to the claims that may be released. This design turns auditability from a documentation aspiration into a verifiable property, and it is what allows the study to report positive effectiveness evidence with precisely scoped interpretation: detector scores retained ranking, the structured case representation ranked well and was expert-validated, numerical routing recovered many reader misses, and ACSC converted that recovery into a safe, high-coverage operating point.

The audit trail is what makes these results credible and reusable. Source-role non-interference prevents query truth from becoming a tuning feature. Historical non-assertion preserves useful cases without treating their outcomes as current facts. Immutable reader records allow later routing to be inspected without attributing the route to the generator. Claim admissibility ensures that each released claim is tied to a passed gate, so the evidence base can be extended or re-audited without re-running the full pipeline.

4.2. Expanded Calibration Turns Ranking into a Usable Operating Point

The HAI results show how the protocol converts continuous detector evidence into a calibrated disposition. TranAD detected every attack episode, the ensemble AUPRC was high, and the continuous D-score preserved event ordering. The protocol's value is that it separates this retained ranking from the categorical operating point and then calibrates the latter rather than asserting it.

The original $0/4$ negative specificity had an extremely wide exact interval, reflecting an underpowered denominator rather than a precise estimate of zero. The expanded 32-negative calibration resolved this: the D-score AUROC rose to 0.902 and the operating point attained a negative specificity of $22/32$ with a calibrated abstention region. The protocol thus yields a bounded, abstaining benign disposition—a usable operating point—while correctly reserving unrestricted benign automation, since the residual overlap still admits nuisance errors. Unlike the TEP routing claim, the HAI detector claim has no sealed held-out confirmation: every expanded-frame estimate is in-sample, and a prospectively collected negative stream is the designated next step.

4.3. Structured Retrieval Is an Effective, Validated Primary Channel

The structured QAKB attained the numerically highest graded nDCG@5 in the frozen assay, tied Full RRF on strict top-rank relevance, and supplied both channels for every query. The nDCG@5 advantage over the full hybrid is not statistically distinguished (paired difference -0.0011, 95% CI -0.0140 to 0.0108), so the claim is that the structured channel is at least as effective as the more complex alternatives on this pool, not that it is superior. That a compact structured representation matched more complex retrieval is itself a positive finding: it shows the case-bank design, not representation complexity, drives retrieval quality on this pool. The paired intervals remain compatible with small gains or losses as the memory grows, but the present evidence supports the structured channel as the effective primary retriever.

Embedding removed the few broad top-rank misses and so serves as a useful secondary discovery channel; promoting it to primary would require a frozen prospective case pool and a practically meaningful paired effect. The phase-PCA embedding result also does not generalize to a complete TS-RAG encoder implementation, which we note as a boundary rather than a limitation of the structured channel itself.

4.4. Expert Validation Grounds the Retrieval Reference

The automated assay completed every pair and passed its synthetic controls, establishing execution integrity. Because repeated calls to one model are not independent experts, and because language-model judges exhibit known position and verbosity biases [32,33], the protocol prespecified a blinded human check rather than resting on the assay alone—and that check confirmed the reference. On the 160-pair expert subset the two domain experts reached substantial agreement (weighted kappa 0.684, meeting the prespecified target of at least 0.60), the assay tracked expert judgment with moderate positive agreement (weighted kappa 0.547), and the expert-graded QAKB nDCG@5 of 0.918 remained high. Because the expert subset graded the structured channel's pairs, this validates the assay-to-human alignment and the structured channel's quality, rather than a cross-method re-ranking under expert grades. The retrieval claim is therefore grounded in human relevance, not only an automated assay, and the independent retrieval sample remains the 49 queries rather than the 1225 pair grades. Because both graders are co-authors or collaborators, familiarity with the case bank could still shape grading despite the blinded protocol; an independent third-party grading round would further strengthen the reference.

4.5. ACSC Delivers Safe, High-Coverage Disposition

Moving disposition from the original reader to a numerical router recovered many fault runs on the exposed stability cohort: the Training-fitted router recognized 330/400 faults. ACSC then converted that recovery into a safe operating point: on the selection partition used for candidate screening it reduced fault-to-benign errors to 6/400 and normal-to-fault errors to 4/100 while meeting both prespecified one-sided risk bounds and attaining the highest admissible coverage among the compared methods. The 330/400 stability count and the 6/400 selection-partition count come from different partitions and are not directly comparable. Because ACSC is selected on this same partition, the development bounds are in-sample and post-selection; the sealed confirmation supplies the independent evidence. The two counts—fault recovery and dangerous error—are reported together precisely so the safety–coverage trade-off is explicit and auditable rather than hidden.

The adapted variant used selection-partition out-of-fold information and did not itself pass the full gate, so the protocol correctly withheld held-out status from it. The single ACSC candidate that passed all development gates was then evaluated once on the sealed confirmation role, where it again satisfied both risk bounds: 9/400 fault-to-benign and 4/100 normal-to-fault raw errors, with one-sided upper bounds of 0.042 and 0.099 respectively, at coverage 0.860. Because the confirmation partition was accessed exactly once, this is an interpretable held-out result—the protocol's discipline is what keeps it interpretable. The normal-to-fault bound passes by a narrow margin: one additional normal-to-fault error on the 100-run sample (5/100) would raise the one-sided 97.5% Clopper–Pearson bound to 0.113, above $\epsilon_{\mathrm{NF}} = 0.10$. The confirmation therefore supports the frozen candidate on this sample while flagging the NF margin as the quantity most sensitive to future data.

4.6. Relationship to Graph, Causal, and Selective Methods

Bayesian-network and graph-neural approaches can model dependencies among process variables and may produce fault classes, posteriors, propagation paths, or root-cause hypotheses [14,15]. They address a different estimand from the present protocol. Our current evidence supports anomaly disposition and evidence lineage, not causal fault localization. Future graph or Bayesian components could be inserted as additional evidence producers only if their source roles and allowed claims were separately recorded.

Selective classification and conformal risk control offer complementary ways to connect coverage and risk [12,13]. The conformal comparator evaluated here satisfied both prespecified bounds but at lower coverage than ACSC, illustrating the coverage cost of a distribution-free guarantee on this frame. The expanded 32-negative set improves alarm-level calibration, although a fully exchangeable, prospectively collected negative cohort would still be needed for a stronger conformal guarantee.

The TEP PCA baseline now has an effect estimate: selecting 36 components by a 95% variance target and thresholding reconstruction error gave a run-level AUROC of 0.876 but exceeded both risk bounds, so it is inadmissible despite reproducible calibration. The same event definitions and data roles apply to the Isolation Forest, quantile, and conformal comparators, keeping the comparison like-for-like; GNN and Bayesian producers would still require separate source-role and claim records.

4.7. Deployment Validation Path

A deployment-oriented program proceeds in five frozen stages; the first four are demonstrated in this work, and the fifth remains the prospective step:

1. expand and freeze an alarm-triggered negative calibration set;

2. obtain a blinded expert relevance subset for the historical case pool;
3. freeze one detector/router configuration and its risk thresholds;
4. evaluate that configuration once on the final confirmation role or a prospectively collected stream; and
5. after dangerous-benign, normal false-fault, coverage, and per-fault gates pass, conduct operator human-factors and plant shadow-mode evaluation.

Automatic control action remains outside this path until independent risk gates pass. Shadow-mode evaluation should measure workload, response time, confirmation bias, distribution shift, and downstream consequences without allowing the system to actuate the plant; established frameworks on automation misuse and disuse [37] and situation awareness [38] inform the design of these measures.

4.8. Scope Boundaries and Future Work

The protocol's gates make the remaining scope explicit rather than hidden, and each boundary points to a concrete next step. The expanded 32-negative HAI cohort substantially narrows specificity uncertainty; a prospectively collected negative stream would tighten it further and would also support a stronger conformal guarantee. The retrieval reference is now human-validated on a 160-pair subset, and the 25-case memory—stress-tested to 200 cases without material reversal—is ready to scale as more cases accrue. The reader diagnostic (32 events) isolates context sensitivity, and the end-to-end claim rests instead on the detector, retrieval, routing, and confirmation gates. The TEP confirmation is a single held-out evaluation of one frozen candidate on a public benchmark by design; that discipline is exactly what makes it interpretable, and a prospective field study is the natural successor. HAI and TEP are public benchmark systems, so operator workload, response time, and control consequences are deferred to the shadow-mode stage. Finally, the protocol targets auditable disposition rather than causal root cause, which graph or Bayesian producers can supply as separately recorded evidence sources.

A cross-cutting limitation is statistical power. The independent denominators are small—49 evaluation events with only four original negative clusters, a 32-event reader cohort, and 100 confirmation normal runs—so exact intervals are wide and single-count changes can move gate outcomes, as the NF margin shows. The protocol responds by reporting exact intervals, separating denominators, and gating claims rather than asserting point estimates; larger prospective cohorts would tighten every bound.

Taken together, the confirmatory experiments close the development loop and demonstrate the protocol's central value: each effectiveness claim is earned by a passed, prespecified gate. The expanded negative calibration turned an underpowered 0/4 into an estimable nuisance distribution; the classical and selective comparison showed that ACSC attains the highest admissible coverage; the expert subset validated the retrieval reference; and the one-shot confirmation reproduced both risk bounds on a sealed partition. The result is an auditable, decision-support diagnosis whose claims are precisely as strong as its evidence—and no stronger.

5. Conclusions

This study formalizes current detector evidence, historical analogies, immutable language explanations, and final dispositions as separate, machine-auditable objects, and introduces audit-constrained selective calibration (ACSC) to maximize coverage subject to simultaneous, prespecified routing-risk bounds. The result is an auditable diagnostic pipeline whose three contributions are each backed by a passed, prespecified gate. First, a label-free median-percentile detector score preserves event ranking and, on an expanded calibration frame, supports a calibrated abstaining disposition; all HAI estimates are in-

frame and in-sample, and only the TEP routing claim rests on a sealed held-out confirmation. Second, structured dual-channel retrieval attains the numerically highest graded nDCG@5 under the frozen assay, not statistically distinguished from the full hybrid, and a blinded two-expert validation confirms that the assay tracks human relevance. Third, numerical routing recovers many faults missed by the language-model reader, and ACSC converts that recovery into the highest admissible coverage among classical and selective comparators while meeting both risk bounds.

Each confirmatory experiment passed its prespecified gate, and the sealed confirmation reproduced both risk bounds (fault-to-benign upper bound 0.042; normal-to-fault upper bound 0.099, the latter by a narrow margin) at 0.860 coverage. The system is decision-support methodology rather than autonomous diagnosis: every released claim is exactly as strong as its evidence. Prospective plant validation, operator human-factors evaluation, and causal localization are the natural next steps.

Supplementary Materials: The supplementary file contains the complete query-bootstrap paired retrieval-difference intervals and the computation detail of the expert-graded nDCG@5. The remaining audit artifacts referenced in the main text—source-ownership manifests, training-normal calibration records, event and case manifests, the 1225-pair assay and control audit, detector-evidence logs, reader/router summaries, prompt, schema, model-identity, and denominator audits, and the failure ledger—are included in the released repository. Raw benchmark data and model weights are not redistributed.

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Institutional Review Board Statement: Not applicable. The study uses public, non-personal HAI and TEP benchmark data and does not intervene in an operating industrial facility. The retrieval-reference validation involved two domain experts grading 160 query–case pairs under a blinded protocol; the experts are co-authors or collaborators of this work, the grading concerned technical relevance of benchmark cases rather than personal data, and no ethics review was required under the authors' institutional policy.

Informed Consent Statement: Not applicable.

Data Availability Statement: HAI 21.03 and the Tennessee Eastman Process are public research datasets [23–25]. The release will center on the analysis code and the derived artifacts it operates on: source-ownership splits, official-label rules, event and run manifests, case versions, hashes, prompts, replay logs, and frozen aggregate results, prepared for release at https://github.com/lplsaiorg/AuditableAlarmDiagnosis_EvidenceRoleSeparation. Raw HAI/TEP data and model weights are not redistributed. The exact repository commit and archival identifier will be recorded when the release is frozen.

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Abbreviations

The following abbreviations are used in this manuscript:

AUPRC	Area under the precision–recall curve
AUROC	Area under the receiver-operating-characteristic curve
ACSC	Audit-constrained selective calibration
CBR	Case-based reasoning
CI	Confidence interval
DTW	Dynamic time warping
HAI	HAI 21.03 industrial control system dataset
LLM	Large language model
MRR	Mean reciprocal rank
nDCG	Normalized discounted cumulative gain
PCA	Principal component analysis
QAKB	Quality-aware knowledge base
RAG	Retrieval-augmented generation
RRF	Reciprocal rank fusion
TEP	Tennessee Eastman Process

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